



GeoGliders × bp

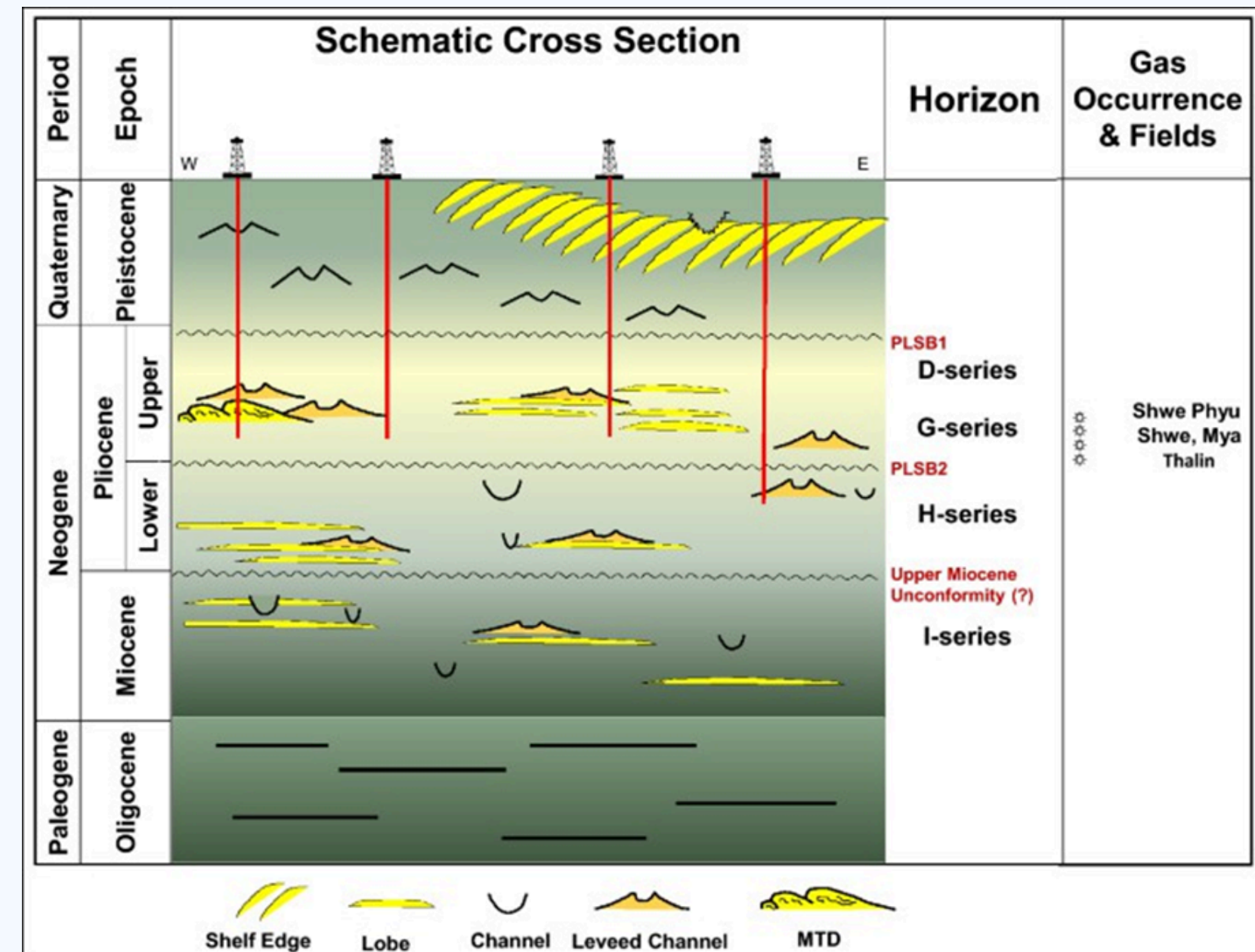
Integrated Subsurface Case Study & Report

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Field Geology & Petroleum System Elements

Field Overview

- Pliocene–Recent turbidite sedimentation dominates
- Deep-water passive margin basin setting
- Thermal subsidence + gravity-driven faulting controls basin evolution
- Growth faults & rollover structures evident on seismic
- Thick marine shale sequences deposited throughout



Field Geology & Petroleum System Elements

SOURCE ROCK

- Jurassic marine shale at 3000–3200 m
- High thermal maturity (thermogenic)
- Type II/III organic matter

RESERVOIR

- Pliocene turbiditic sandstones
- Porosity $\approx$ 0.20–0.25 (Alpha-1)
- High-amplitude continuous reflectors

SEAL

- Thick regional marine shale
- Low acoustic impedance contrast
- Minor risk along fault planes

TRAP

- Combination structural-stratigraphic
- Fault-bounded rollover closures
- Confirmed on PSDM seismic

MIGRATION

- Vertical migration along faults · Confirmed gas in Alpha-1 & Ruby · Flat spots on seismic

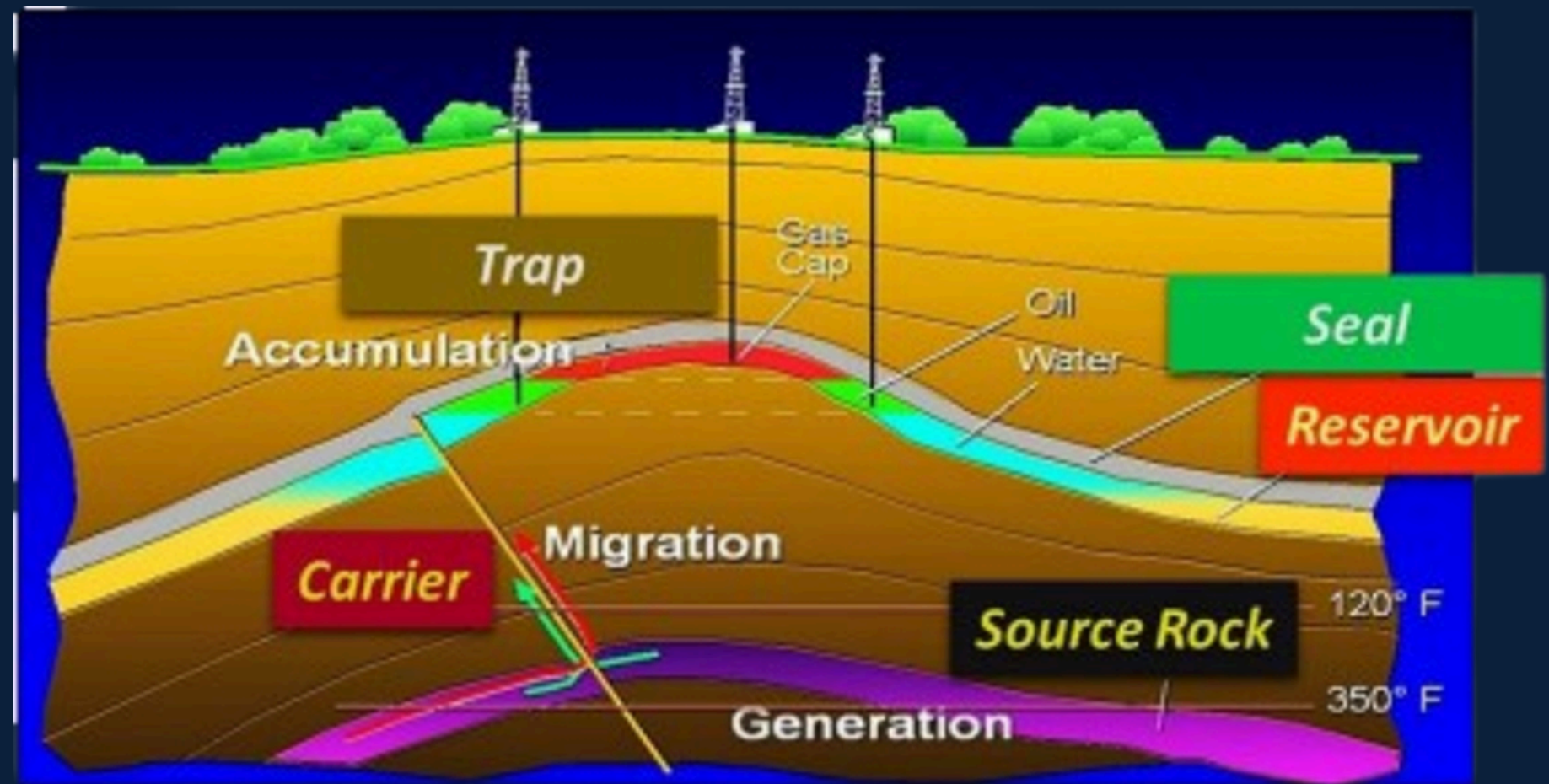
Thermogenic Gas System – Mustang Basin

SOURCE TYPE

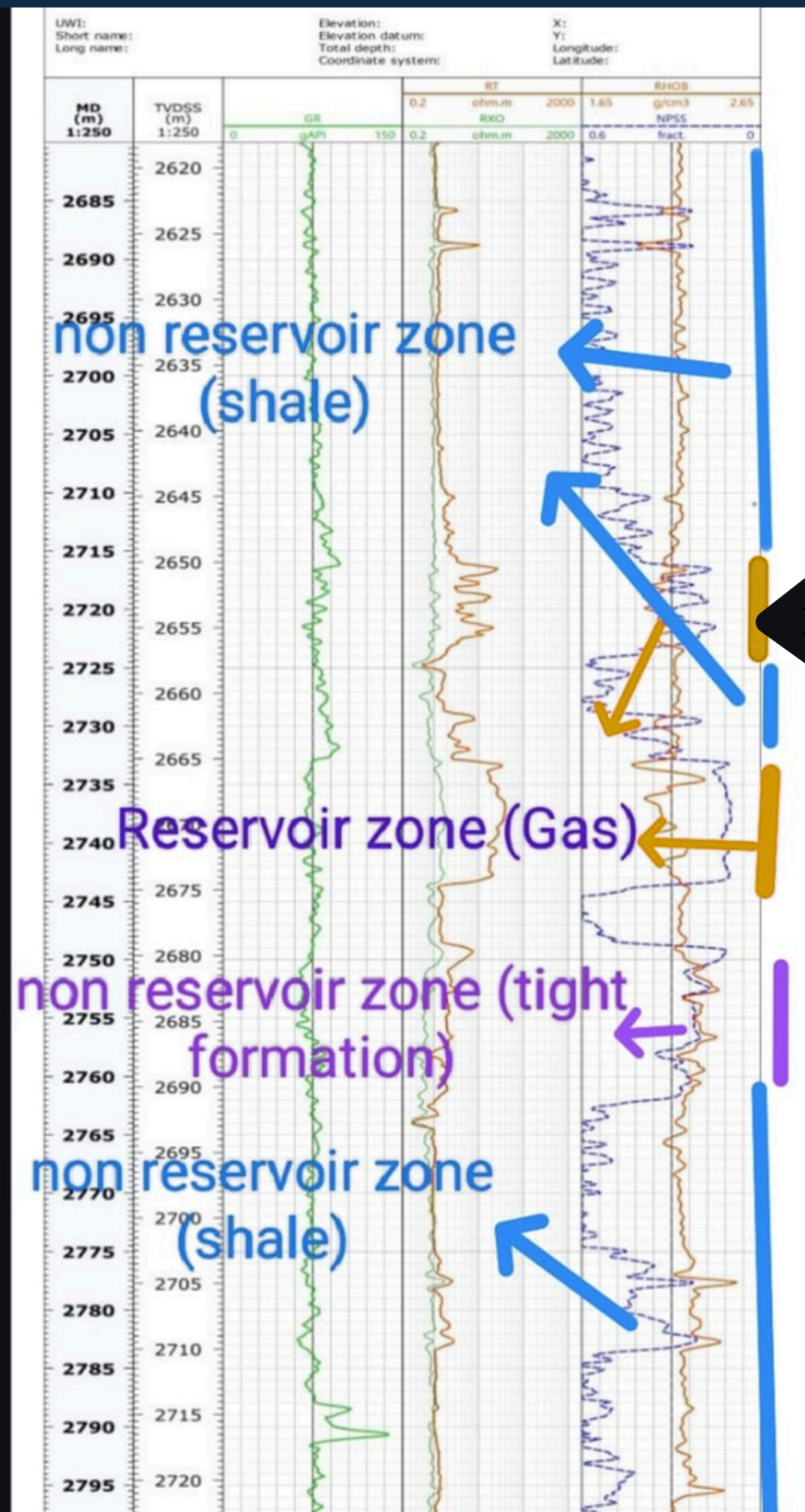
- Thermogenic marine source rock
- Gas-prone – high thermal maturity
- Organic matter Type II/III (marine–mixed)

HYDROCARBON PHASE

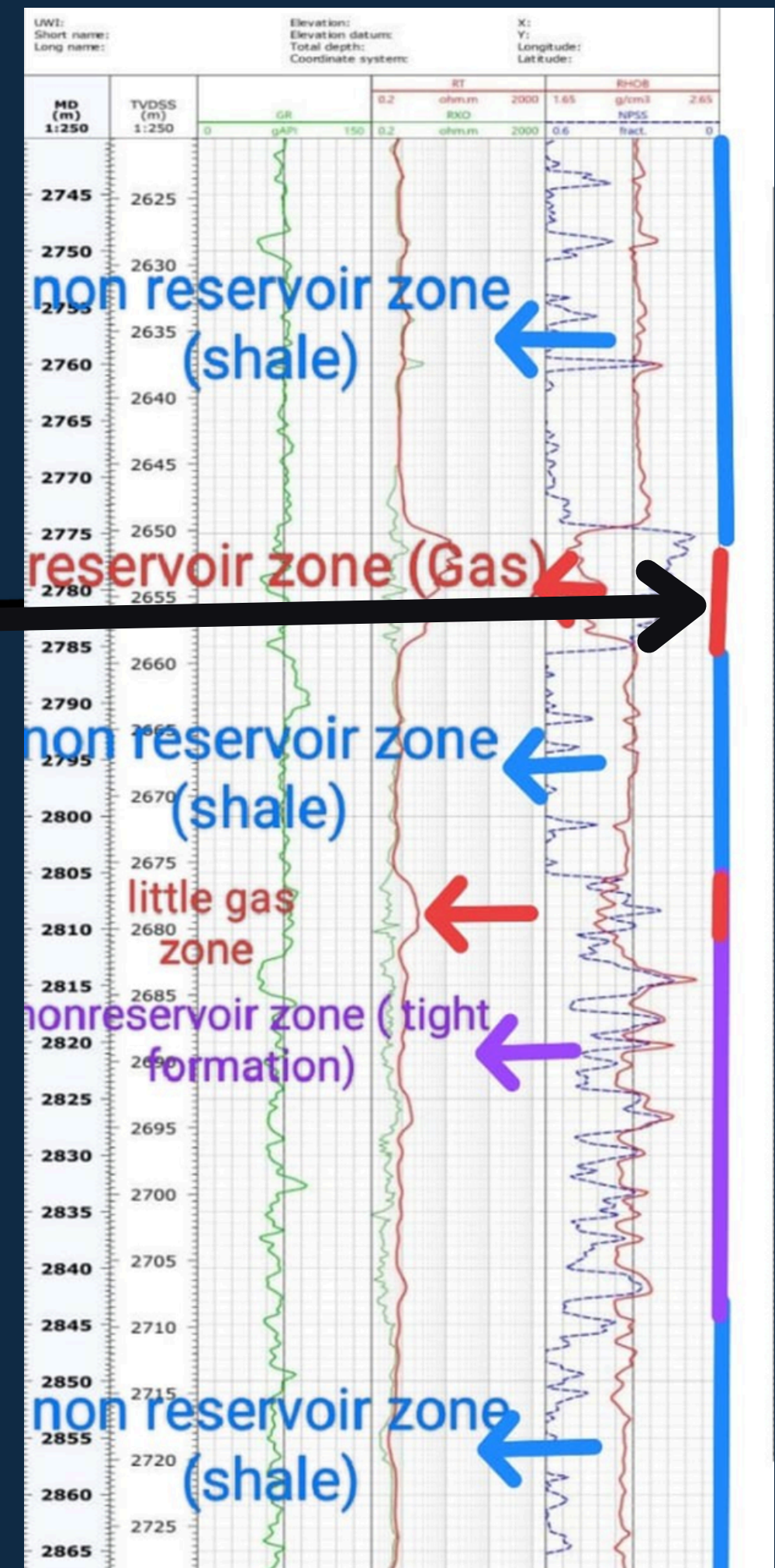
- Predominantly dry gas confirmed
- Gas gradient from MDT pressure data
- High-rate gas flow in DST tests
- No oil shows detected in cores/logs
- Deep burial limits biogenic gas probability

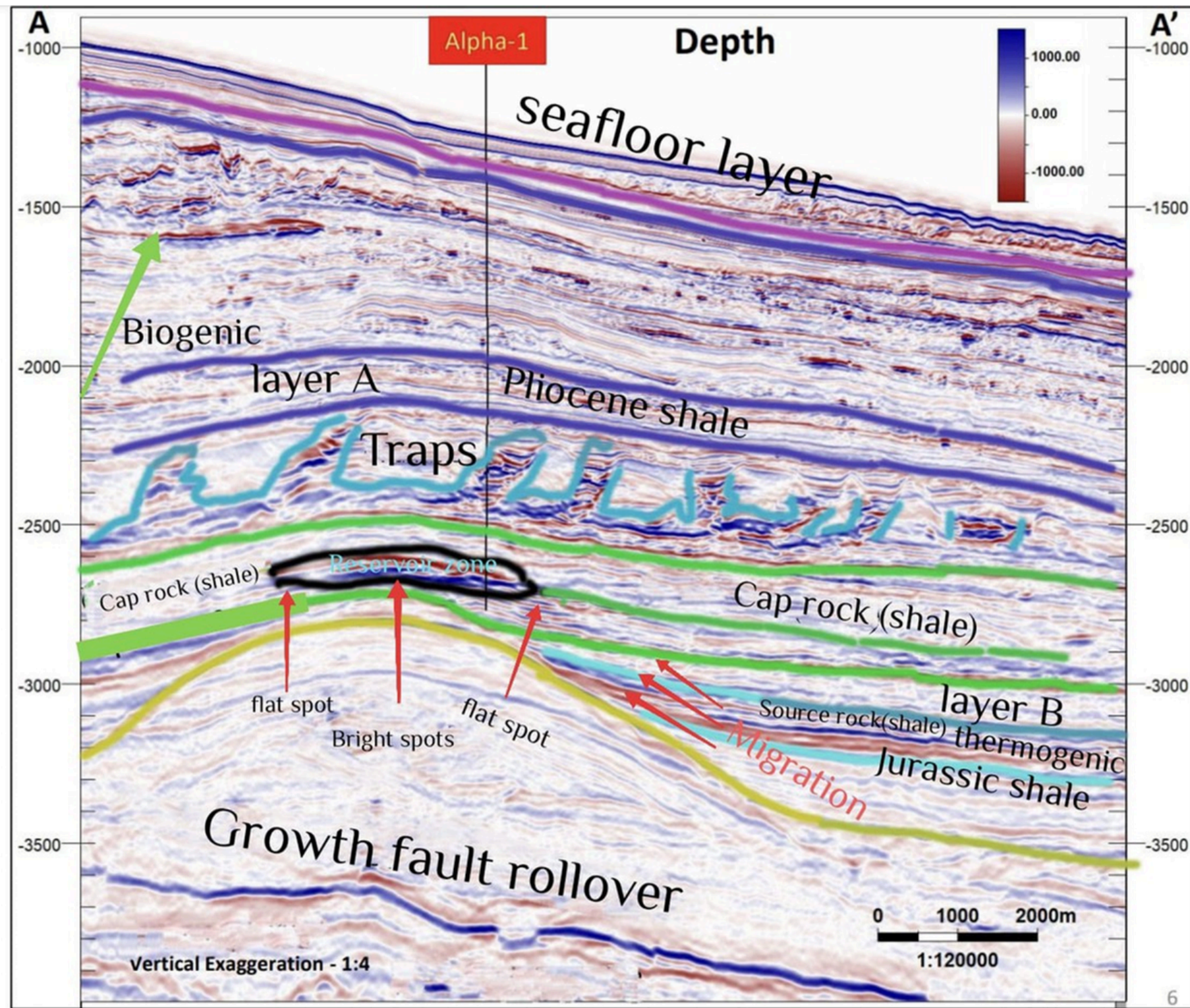


Identify possible source type and phase of HC
Already shown in Seismic Section



Well Tie shows reservoir homogeneous reservoir





Source Rock

- Jurassic shale (~3000–3200 m)
- Thermally mature → generates thermogenic hydrocarbons

Migration

- Jurassic shale (~3000–3200 m)
- Thermally mature → generates thermogenic hydrocarbons

Main Reservoir (2665–2675 m)

- Identified by seismic + RMS amplitude + Alpha-1 logs
- Bright spot anomaly → hydrocarbon-charged sands
- Sealed by overlying shale → effective trap

Secondary Reservoir (2340–2430 m)

- Channelized sand bodies
- Stratigraphic trap
- Sealed by Pliocene shale

Shallow Gas Zone (1250–1450 m)

- Amplitude anomalies
- Biogenic gas accumulations
- Likely self-sourced (minimal migration)

Geological Setting

- Passive-margin (Mustang Basin)
- Growth faults consistent with basin evolution

Deep Structure (>3000 m)

- Growth-fault rollover anticline
- Syn-sedimentary origin
- Provides migration pathways + structural closure

Think about different processing types and methods and suggest if they can be used to improve the image of Seismic Section.

WHY NEEDED

- Fault zones → mis-positioning
- Reflector discontinuity below faults
- Residual moveout in gathers

What can be done

1. Velocity Model Updating (Tomography)

- Improves depth positioning, reduces reflector mis-tie, and sharpens fault images.

2. Anisotropic PSDM (TTI migration)

- Corrects imaging of dipping reflectors and faults in anisotropic shales.

3. Pre-stack conditioning (true amplitude processing)

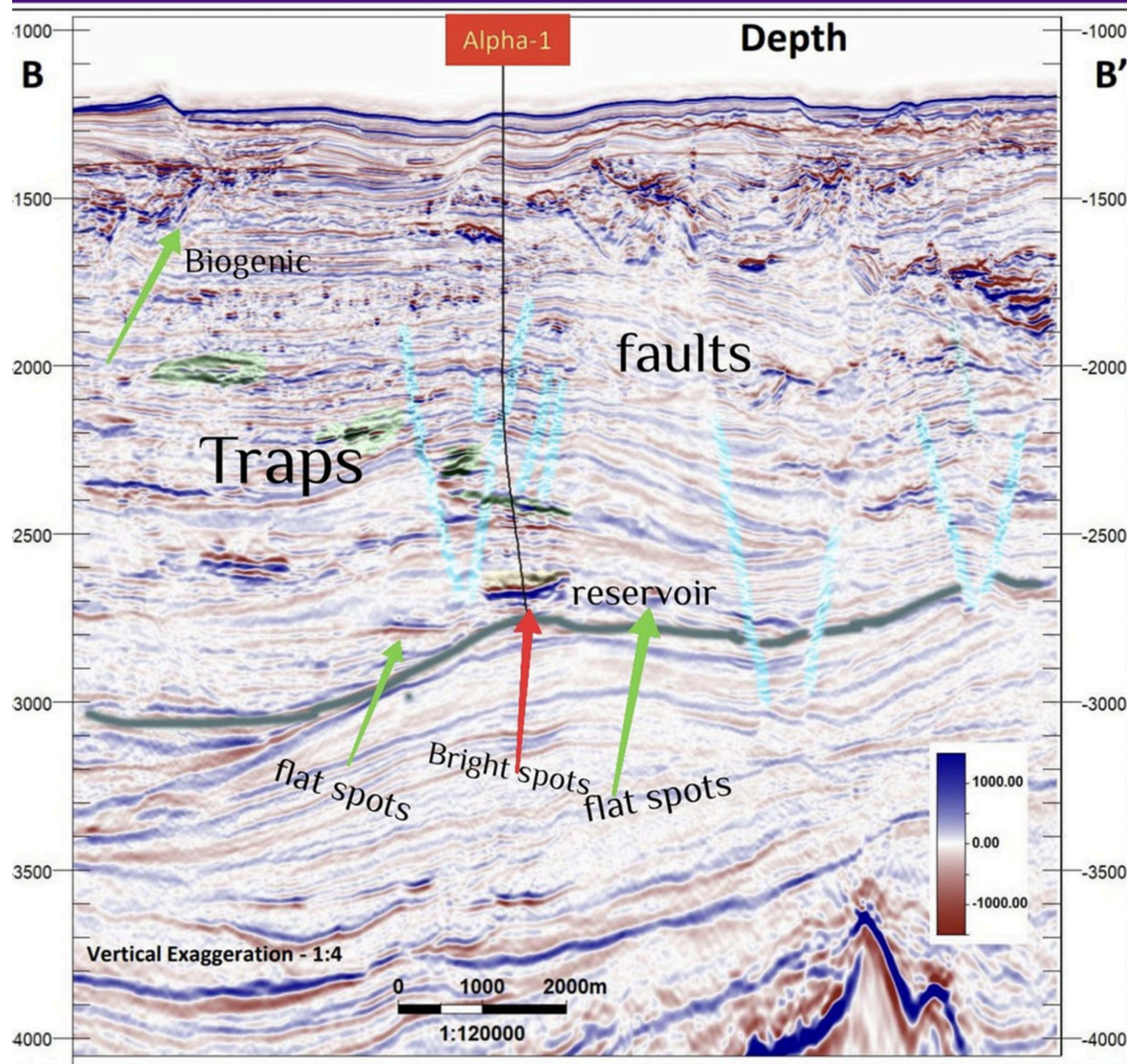
- → Preserves amplitudes and improves reliability of AVO and DHI interpretation.

4. Angle-domain common image gather (ADCIG) analysis

- Identifies residual moveout and helps refine velocity models for better migration.

5. Inversion & Attributes

- Enhances reservoir characterization and improves sand–shale discrimination.



1.Data & Processing

- 3D seismic data processed using PSDM

2.Seafloor Indicator

- Strong, continuous top reflector → seafloor

3.Structural Features

- Multiple normal faults (mid to deep section)
- Faults act as migration pathways

4.Reservoir Interpretation

- Probable reservoir zones in channelized sands
- Sands sealed by overlying layers → hydrocarbon trapping

5.Impedance Feature

- Low-impedance zone in middle section

6.Well Tie

- Alpha-1 well penetrates crestal part of reservoir

7.Trap Type

- Predominantly combination traps

8.Seismic Character

- Uniform polarity

TWT at Alpha-1well

TVDSS Base (m)	Interval TWT (ms)	Cumulative TWT (ms)			
2622.5	1.34	1.34			
2623.0	0.26	1.60			
2625.0	1.06	2.66			
2627.0	1.10	3.76	TVDSS Base (m)	Interval TWT (ms)	Cumulative TWT (ms)
2653.0	13.88	17.64	2710.0	2.66	48.12
2655.0	1.10	18.74	2720.0	5.34	53.46
2665.0	5.34	24.08	2721.0	0.52	53.98
2675.0	5.54	29.62			
2680.0	2.66	32.28			
2690.0	5.18	37.46			
2705.0	↓	45.46			

Petrophysical and Pressure Data Analysis

Q1. & 2 Identify the reservoir and non-reservoir intervals in the well log for all three wells.

Density porosity of gas is high and neutron porosity is low, so from neutron–density crossover we define gas zones in well logs

as(units are md)

ALPHA-1

GAS

2715 m – 2725 m

2730 m – 2744 m

tight rock:

2750 – 2760 m

Shale:

2685 – 2715 m

2795 – 2730 m

2745 – 2750 m

2760 – 2795 m

RUBY

Shale:

2745 – 2775 m

2785 – 2805 m

2845 – 2885 m

Gas:

2775 – 2785 m

Very dense rock

(compacted / hydrocarbon):

2805 – 2845 m

(From MDT plot we get a deposit = 3.365 g/cc

in this area which is extremely high from normal case

NOVA

$\rho_b \approx 2.15 \text{ g/cm}^3$

Which also confirms that it is all shale.

Shale:

2540 m – 2615 m

Q3. Comment on the reservoir quality and nature of the sands MDT confirms good vertical connectivity

Mostly shale

Q4. Identify the type of fluid present in the reservoir (oil, gas, or water). Justify your answer using log and MDT evidence

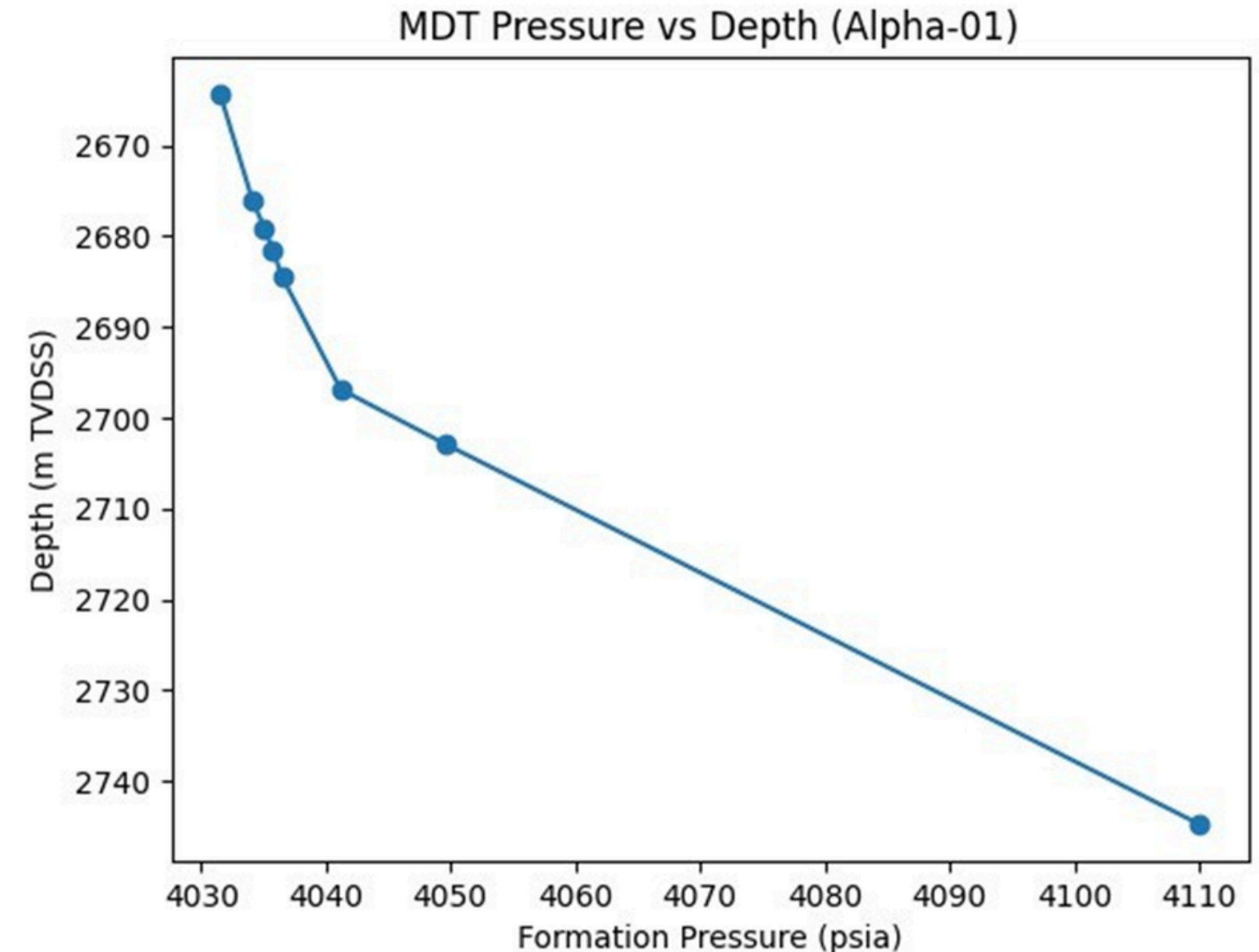
Formation Pressure Data for Alpha-1

Upper Interval:

- 2664 – 2697 m TVDSS
- Gradients are (0.22 – 0.38 psi/m) Indicates gas bearing column
- $\rho \cdot g = \text{slope (psi/m)}$
- Slope = 0.23 psi/m
- $\rho = 1517 / 9.81 = 155 \text{ kg/m}^3$
- (Slope = 0.38 psi/m)
- $\rho = 2620 / 9.81 = 267 \text{ kg/m}^3$
- So, the density falls in typical gas fluid density range, hence verified.

Lower Interval

- (> 2697 m) TVDSS
- Pressure gradient:
- 1.4 psi/m
- $\rho = 1.4 \cdot 6894.76 / 9.81 = 984 \text{ kg/m}^3$
- It corresponds to a Water Zone (formation brine).



The above result is also justified by the log data as shown in part 1 of this question

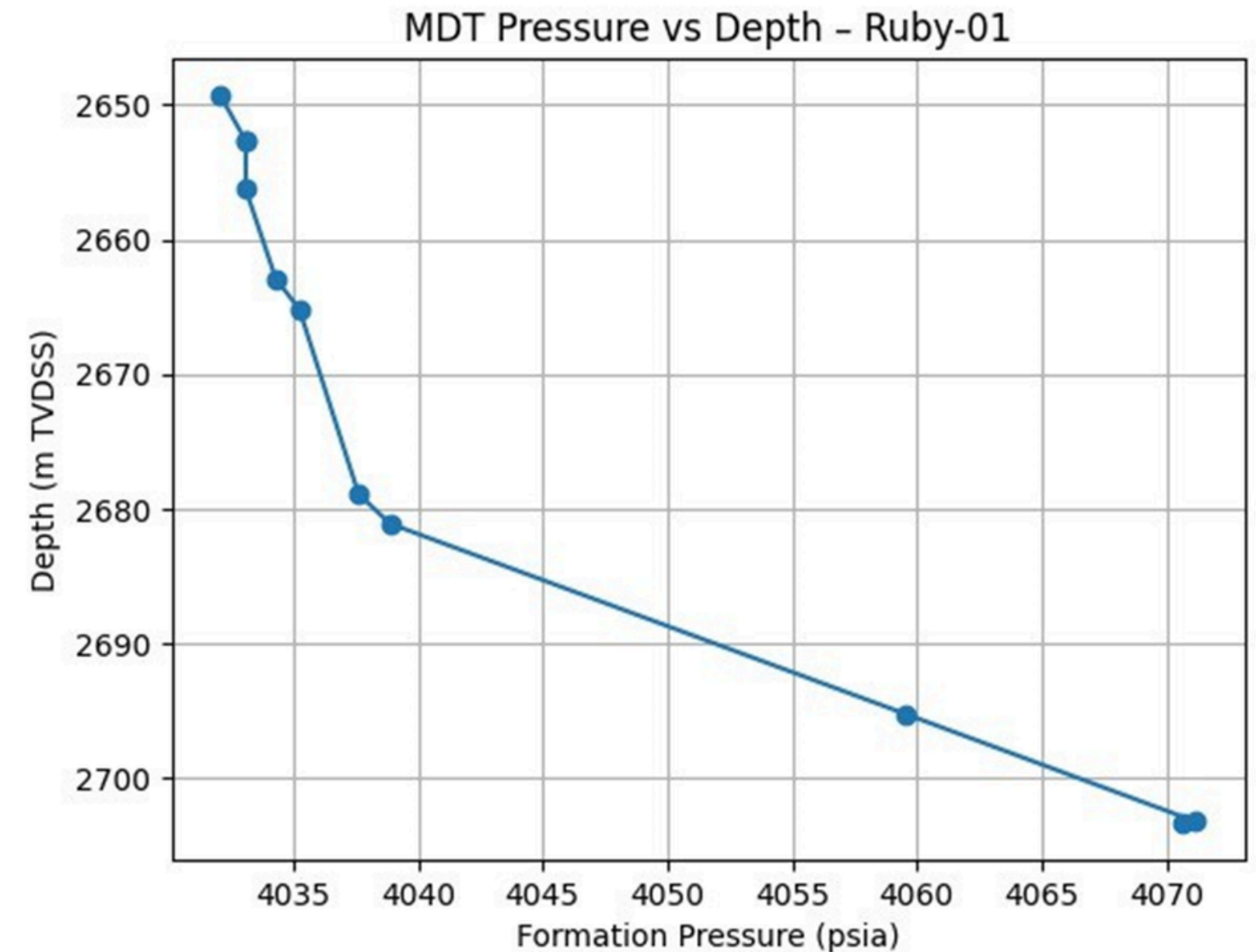
FormationPressureDatafor Ruby

UPPER INTERVAL

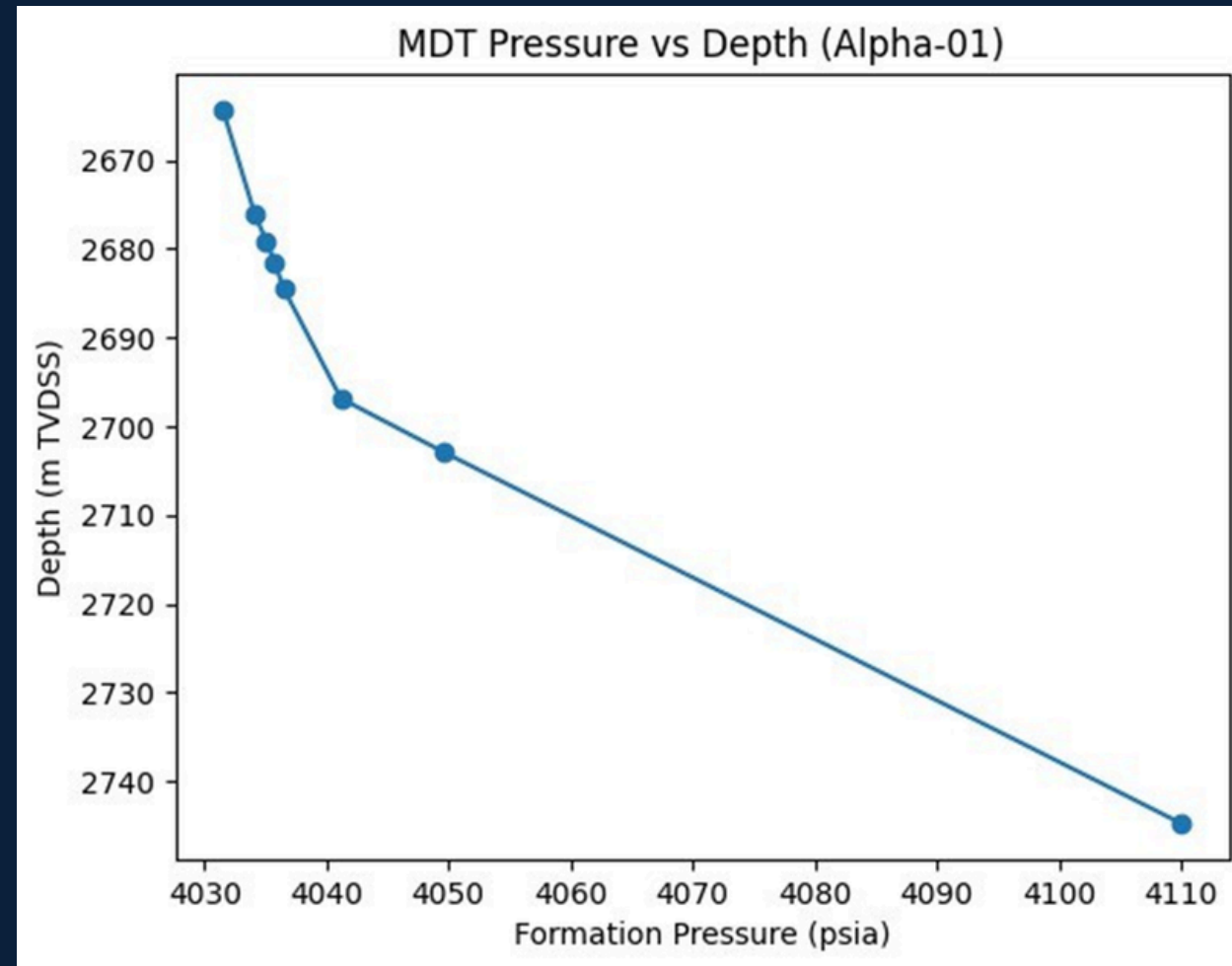
- (2649 – 2681 m TVDSS):
- Gradients = 0.05 – 0.17 psi/ft Mostly gas-type gradients, i.e. 7.0 – 9.7 psi
- Mostly gas-condensate zone

LOWER INTERVAL

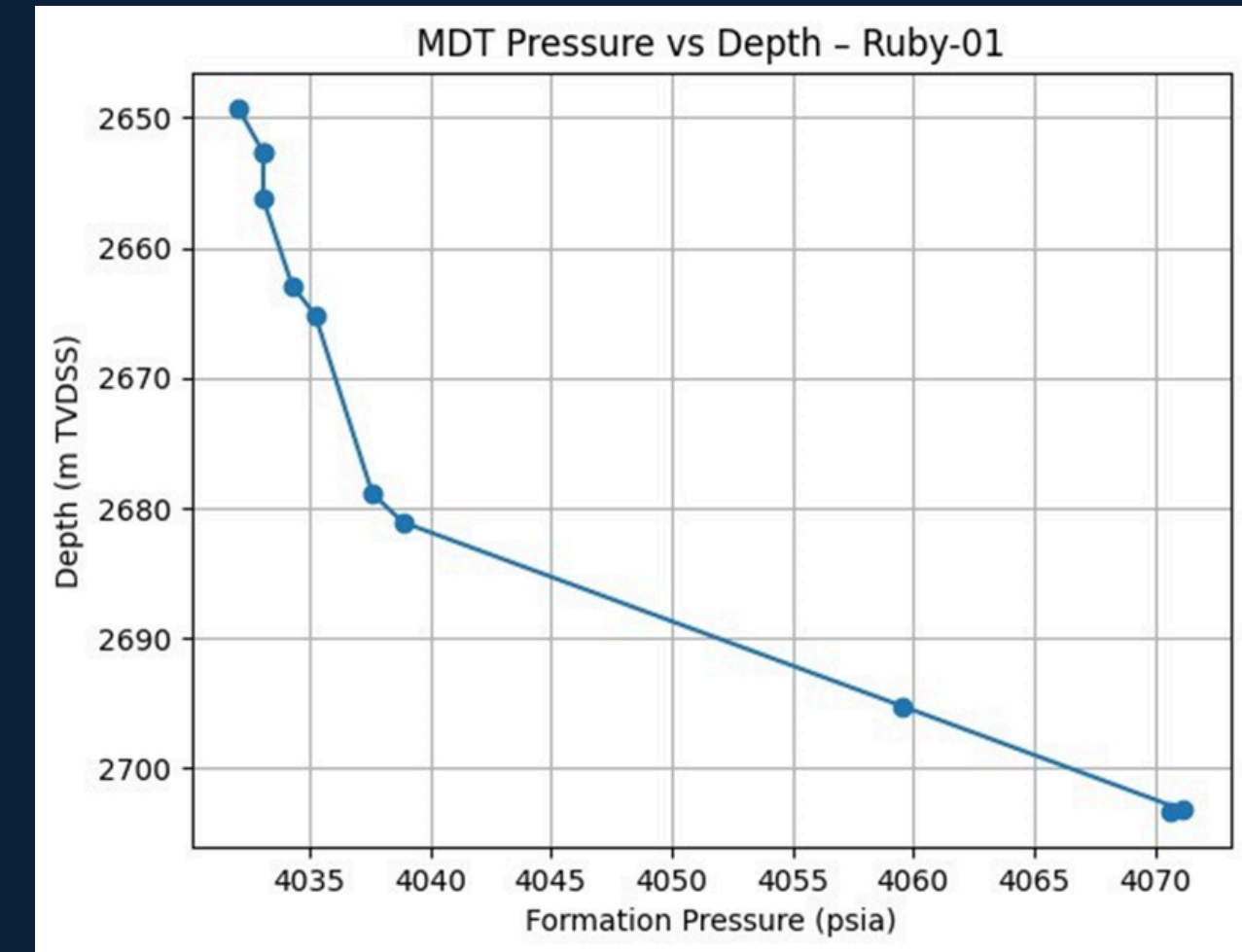
- (2681 – 2973 m TVDSS):
- Gradient $\approx$ 0.45 psi/ft
- → It shows a water leg below



Q5. Identify the water contact (GWC or OWC) from log data and support your interpretation using MDT pressure data



Alpha-01:
GWC = 2697 m (TVDSS)
Ruby-01:
GWC lies between 2681
– 2845 m TVDSS



Q6. Calculate the average porosity & water saturation of the hydrocarbon-bearing zone

Ruby-01:

Average fluid density from MDT plot $\approx 0.34 \text{ g/cm}^3$

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

Where:

- ρ_b (from log) = 2.15 g/cm³
- $\rho_{ma} = 2.65$ g/cm³

$$\phi = \frac{2.65 - 2.15}{2.65 - 0.34} = \frac{0.5}{2.31} \approx 0.22$$

$$S_w = \sqrt{\frac{0.1}{(0.22)^2 \times 20}} = 0.387$$

Alpha-1

Average fluid density from MDT plot ≈ 0.182 g/cm³

Water Saturation:

Porosity Calculation

$$\phi = \frac{\rho_{ma} - \rho_b}{\rho_{ma} - \rho_f}$$

$$\phi = \frac{2.65 - 2.15}{2.65 - 0.182} = \frac{0.5}{2.468} \approx 0.20$$

$$S_w = \sqrt{\frac{0.1}{(0.2)^2 \times 20}} \approx 0.35$$

Q7.Estimate the net pay thickness

Alpha-01: h = 30 m

Ruby-01: h = 10 m

- Uncertainty
- We have not considered small (<5m),
- low porosity zones where small amount of gas might be trapped

Q8.Expected permeability of the reservoir

(Alpha-1)

$$\begin{aligned} V_{sh} &= \frac{GR_{log} - GR_{clean}}{GR_{shale} - GR_{clean}} \\ &= \frac{75 - 50}{135 - 50} \\ &= 0.294 \end{aligned}$$

$S_{wi} = 15\%$

$$\begin{aligned} \phi_e &= \phi(1 - V_{sh}) \\ &= (0.2)(1 - 0.294) \end{aligned}$$

$$\phi_e = 0.1412$$

$$k = 0.136 \frac{\phi_e^{4.4}}{S_{wi}^2}$$

$$k = 2.4 \text{ mD}$$

- Shaly, laminated fine-grained sands
- High clay content Clay-lined pore throats Poor pore connectivity

Ruby

Effective Porosity = 0.2

Irreducible Water Saturation = 0.387

$$\phi_e^4 = 0.22^4 = 0.00234$$

$$S_{wi}^2 = 0.387^2 \approx 0.1497$$

$$k = 1000 \times \frac{0.00234}{0.1497} \approx \mathbf{15.6 \text{ mD}}$$

- Indicates moderate-quality shaly sand
- Consistent with heterogeneous deep-marine channel deposits
- Adequate for gas flow, but permeability is reduced by shale content and lamination

Fluid Characterization and Volumetric Estimation

1. Identify hydrocarbon fluid type (oil/gas/mixed)

2. Estimate hydrocarbons in place (STOIIP / GIIP) in barrels for oil or bcf for Gas.

- MDT pressures in Alpha-1 and Ruby follow a gas pressure gradient DST results show very high flow rates
- Provided $B_g = 0.004$ (gas FVF), which is explicitly meant for gas volumetrics
- Therefore, the field is classified as a gas field
- GIIP Formula (Standard Volumetric Method)

$$GIIP = \frac{7758 \times A \times h \times \phi \times (1 - S_w)}{B_g}$$

Where:

- A = Area (acres)
- h = Net pay thickness (ft)
- ϕ = Average porosity
- S_w = Water saturation
- $B_g = 0.004$ (given in PDF)

- Area = 22 km²
- Net pay thickness $\approx$ 10 m (from logs)
Average porosity $\approx$ 0.20(Alpha-1)
Water saturation $\approx$ 0.35(Alpha-1)
Average porosity $\approx$ 0.22(Ruby) Water saturation $\approx$ 0.387(Ruby)
- $B_g = 0.004$
- 1 km² = 247.1 acres
- 1 m = 3.281 ft

GIIP Calculation (Alpha-1)

$$\text{GIIP} = \frac{22 \times 10^6 \times 30 \times 0.2 \times (1 - 0.35)}{0.004}$$

$$\text{GIIP} \approx 2.15 \times 10^{10} \text{ std m}^3$$

GIIP Calculation (Ruby)

$$= \frac{26 \times 10^6 \times 10 \times 0.22 \times (1 - 0.387)}{0.004}$$

$$\text{GIIP} = 8.77 \text{ BCM}$$

Volumetric estimation shows that Ruby contains higher GIIP than Alpha-1 due to larger area, thicker net pay, and slightly better porosity despite higher water saturation.

Recovery Factor Estimation and Production Forecasting

Calculate Well Deliverability and Absolute Open Flow Potential based on DST

Results.

Well deliverability was evaluated using multirate DST data and fitted with the Rawlins–Schellhardt back pressure equation $p_r=4046.685$ psi (average reservoir pressure from data)

$$q = C(p_r^2 - p_{wf}^2)^n$$

where;

$$\log q = \log C + n \log(\Delta p^2)$$

$$n = 0.709$$

$$C = 0.00359$$

After plotting the graph, Slope = n, Intercept = log(C)

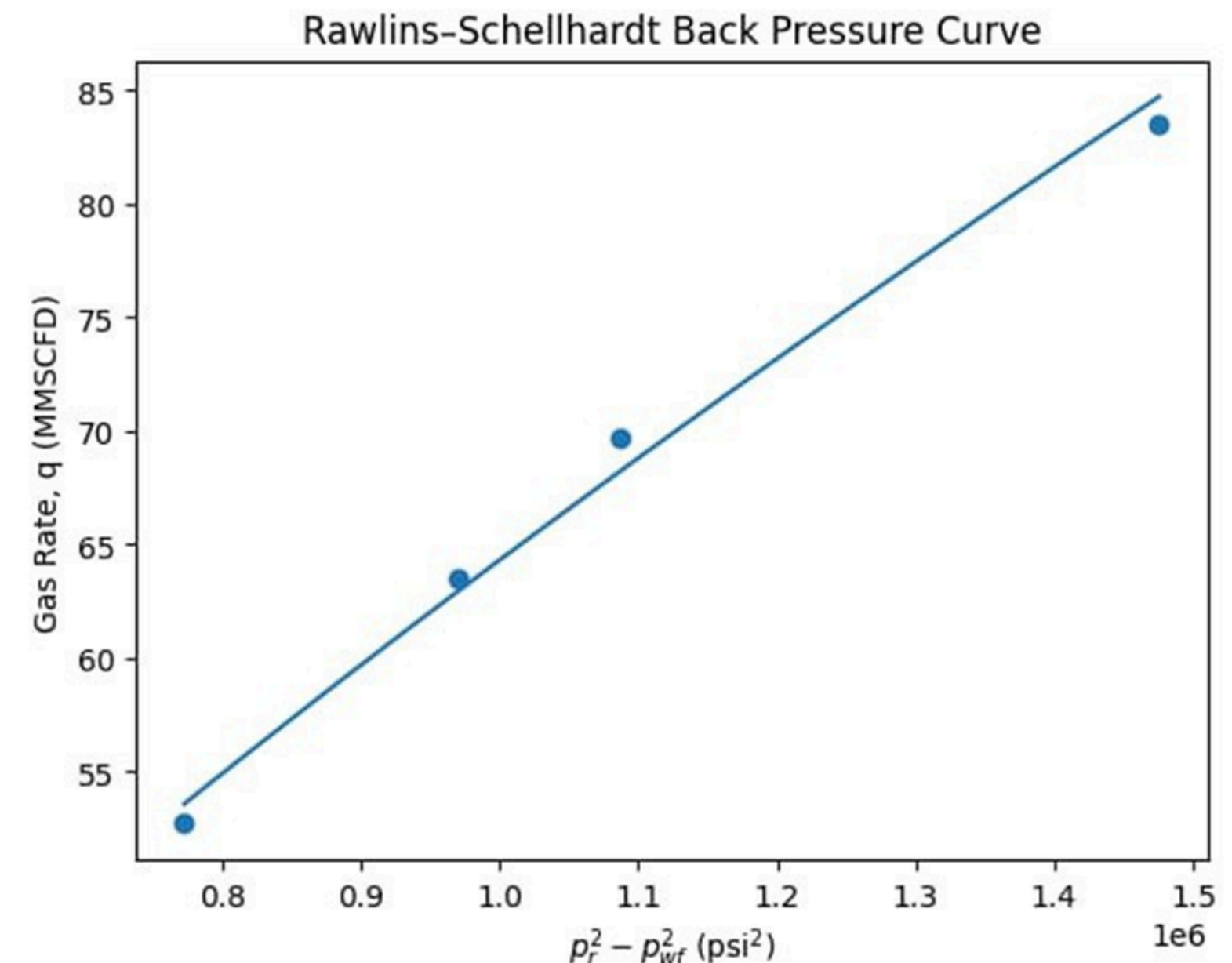
Well deliverability is represented by:

$$q = 3.59 \times 10^{-3} (p_r^2 - p_{wf}^2)^{0.709}$$

$$p_{wf} = 0$$

$$q_{AOF} = C(p_r^2)^n$$

$$q_{AOF} = 467 \text{ MMSCFD}$$



Estimate recoverable hydrocarbon volumes based on available reservoir data and relevant analogues and generate production profiles using decline curve analysis

Initial rate (q_i) = 108 MMSCFD Recovery factor :- 51%

(Median value from given plots)

Observed decline rates :- 12–18% per annum (from given plot)

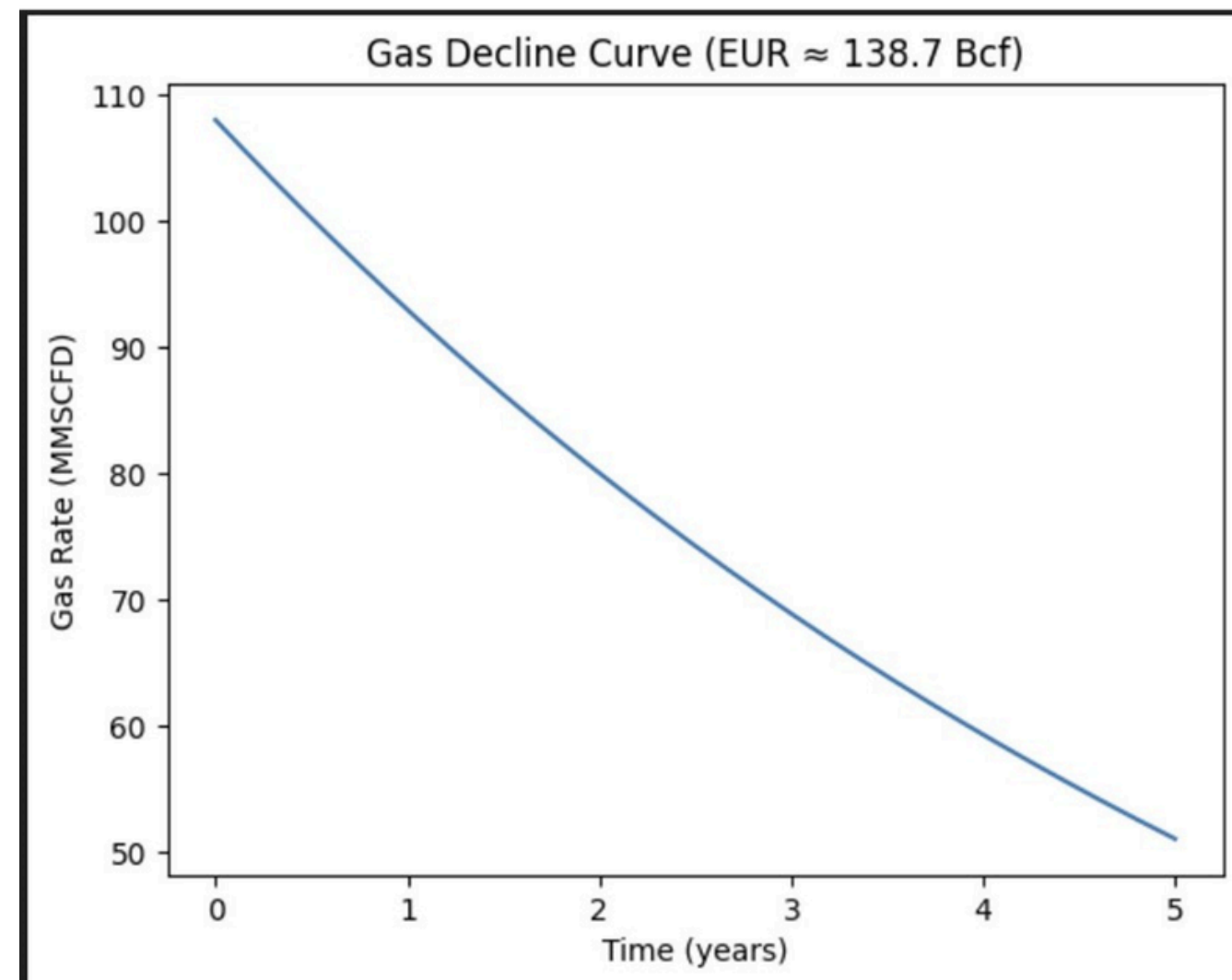
Cumulative Production $\approx$ 138.7 Bcf (EUR)

Cumulative Production ≈ 138.7 Bcf (EUR)

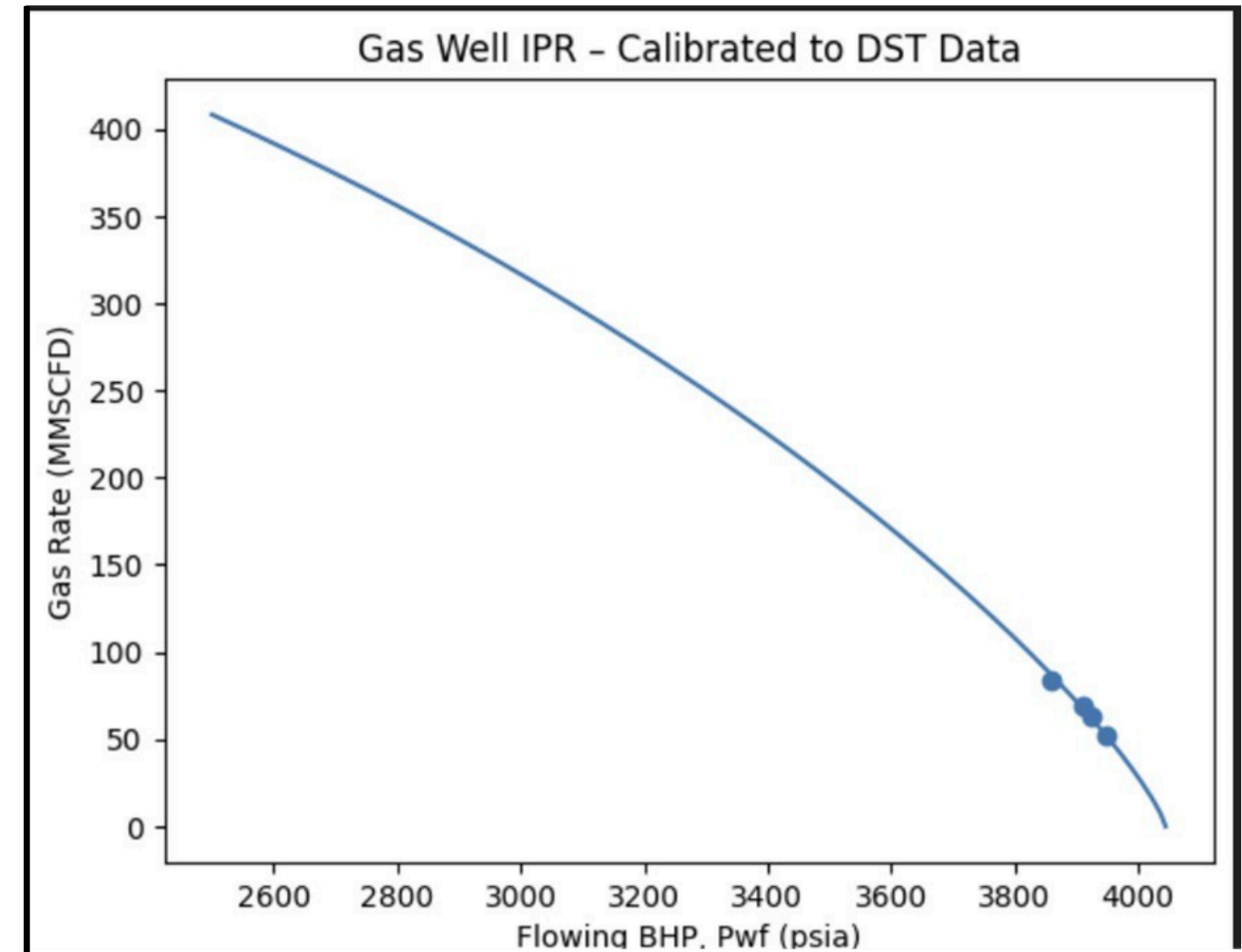
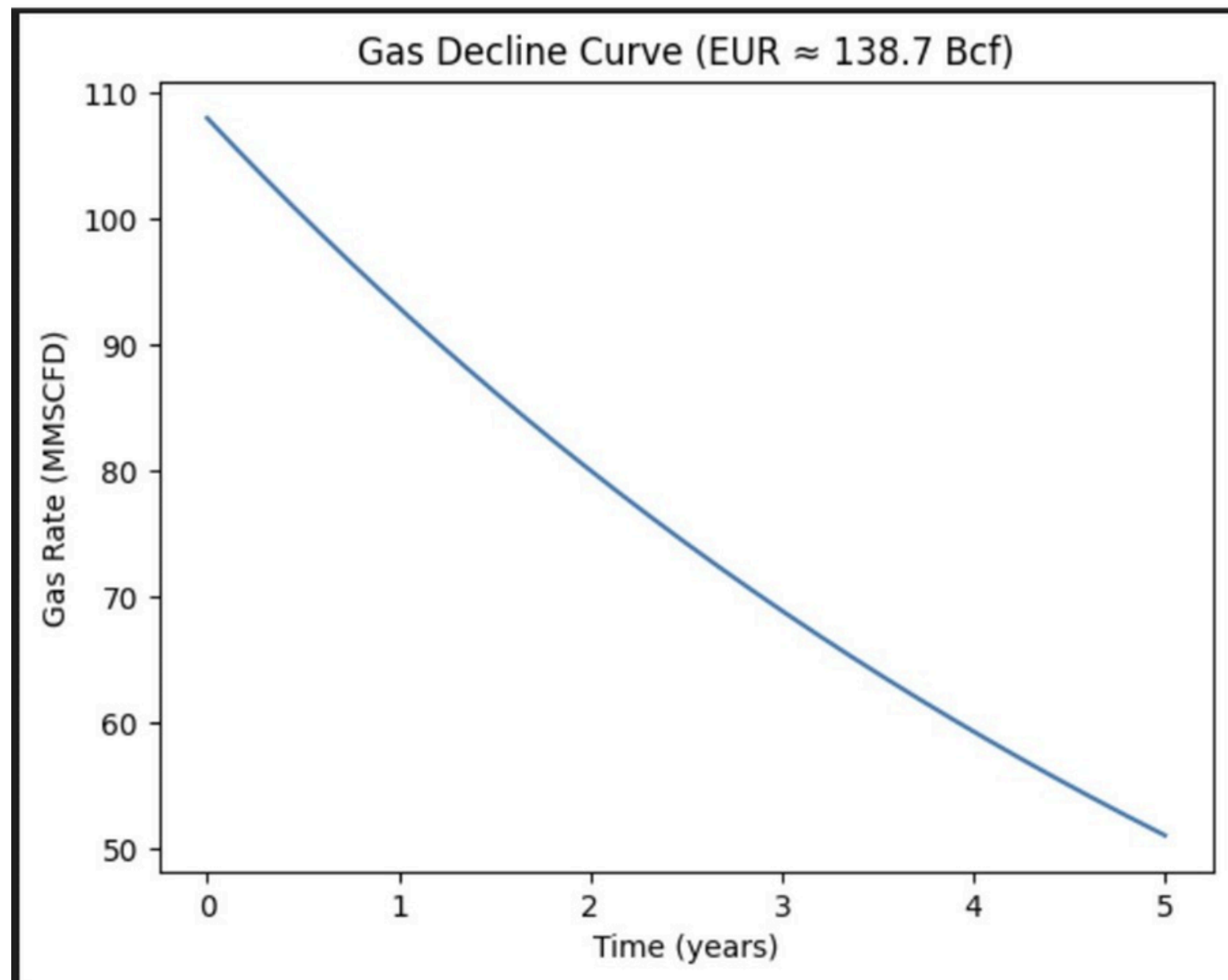
We have assumed Exponential Decline:

$$q(t) = q_i e^{-Dt}$$

$$q_t = 108 e^{-0.12t}$$



Gas Decline Curve & Gas Well IPR



Appraisal and Full-Field Development Planning

Appraisal Strategies:

- Drill one appraisal well near the channel margin (from seismic image) of Alpha-01 to confirm lateral reservoir continuity and facies change.
- Collect core samples in the appraisal wells to better know permeability and saturation by special core analysis directly.
- Get sonic log data to know TWT.

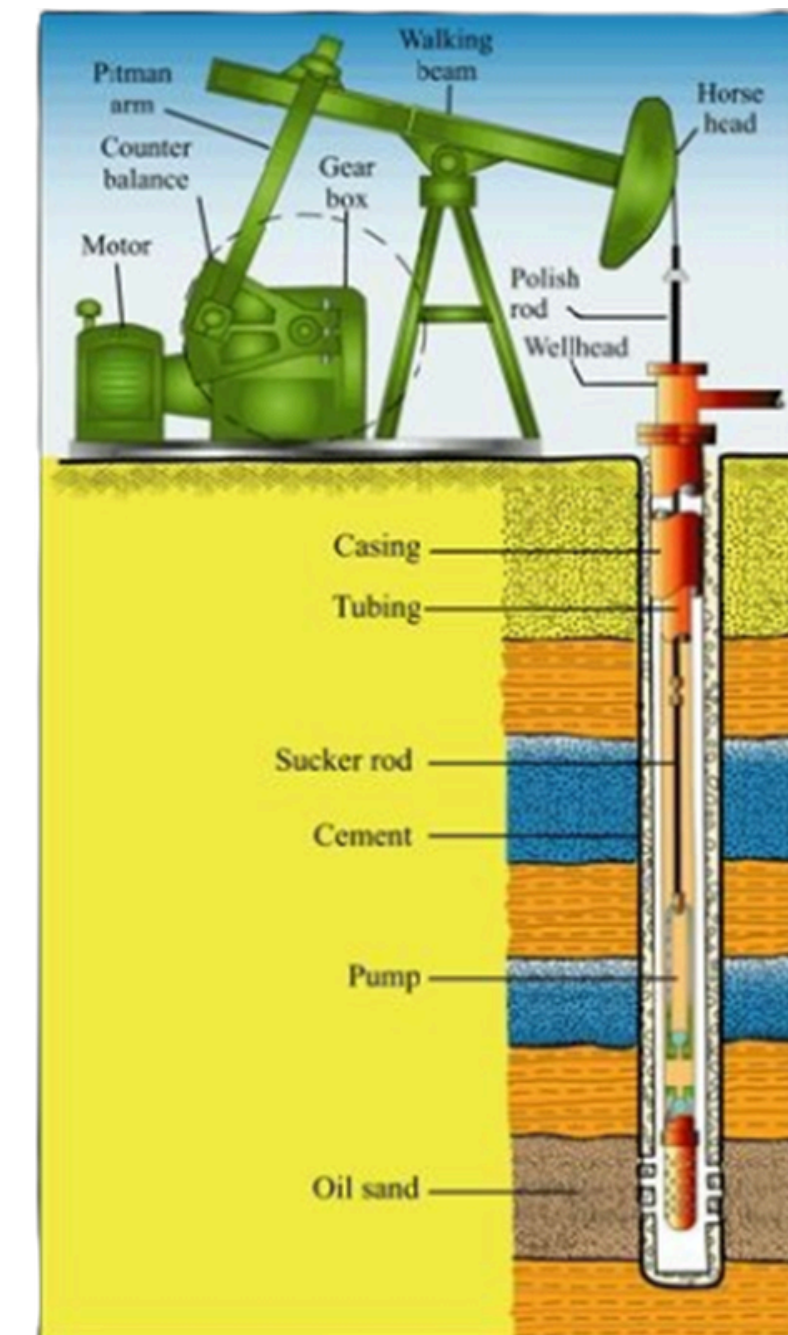
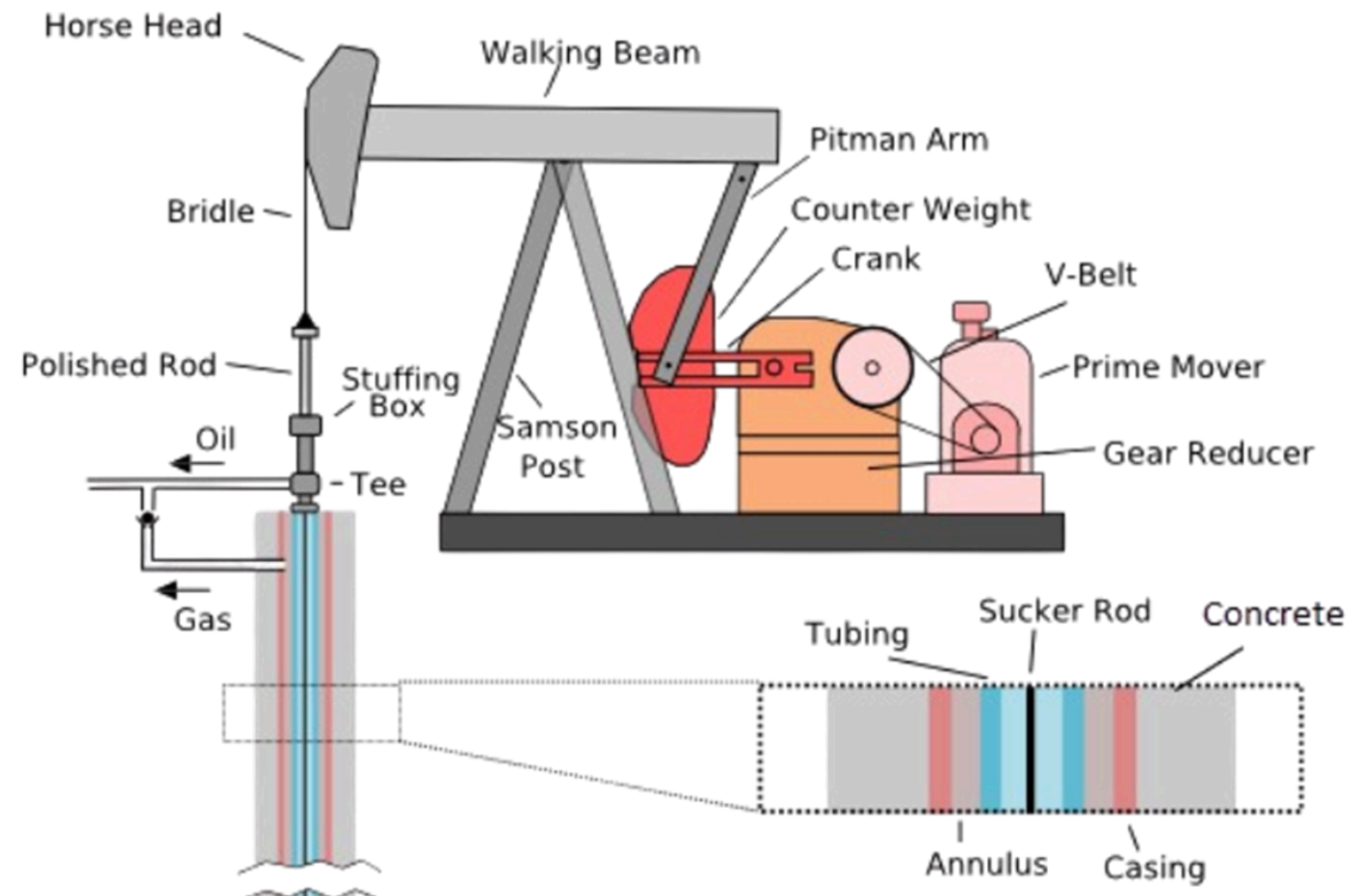
Field Development Plan – Well Spacing

- From the MDT pressure data, we have noticed that there is no indication of strong pressure maintenance.
- So, if the pressure declines rapidly, we can introduce some artificial lift or waterflooding into our wells.
- From the initial production gas rate data, we see that most of the wells like ours are having an initial rate more than 108 MMSCFD, which indicates high well deliverability. This suggests strong gas drive and good reservoir quality, and the need for fewer appraisal wells with wider spacing.

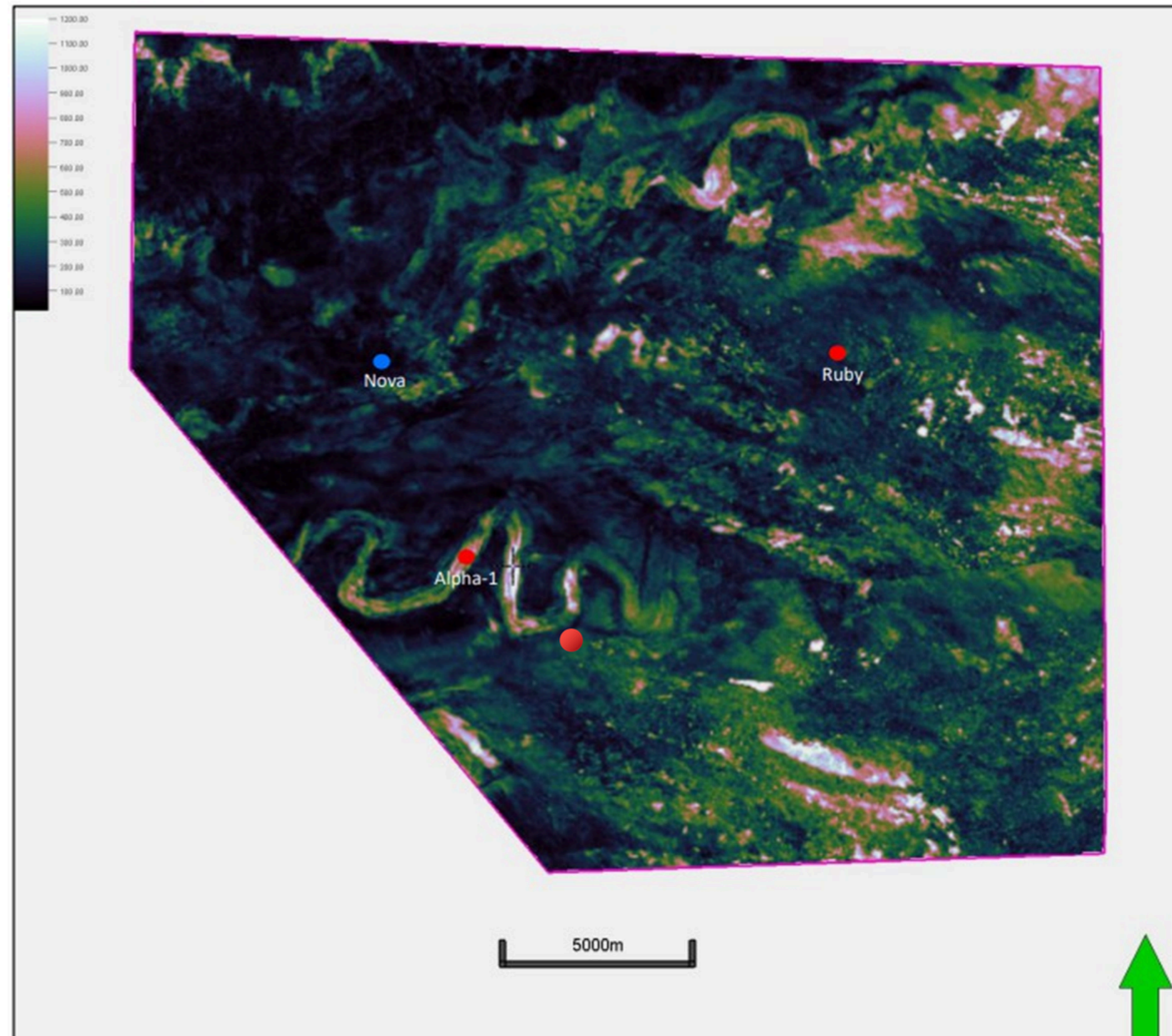
$$\text{Well Count} = \text{EUR target} / \text{EUR per well}$$

Spacing between the wells:

If EUR is higher per well, then wider spacing is needed

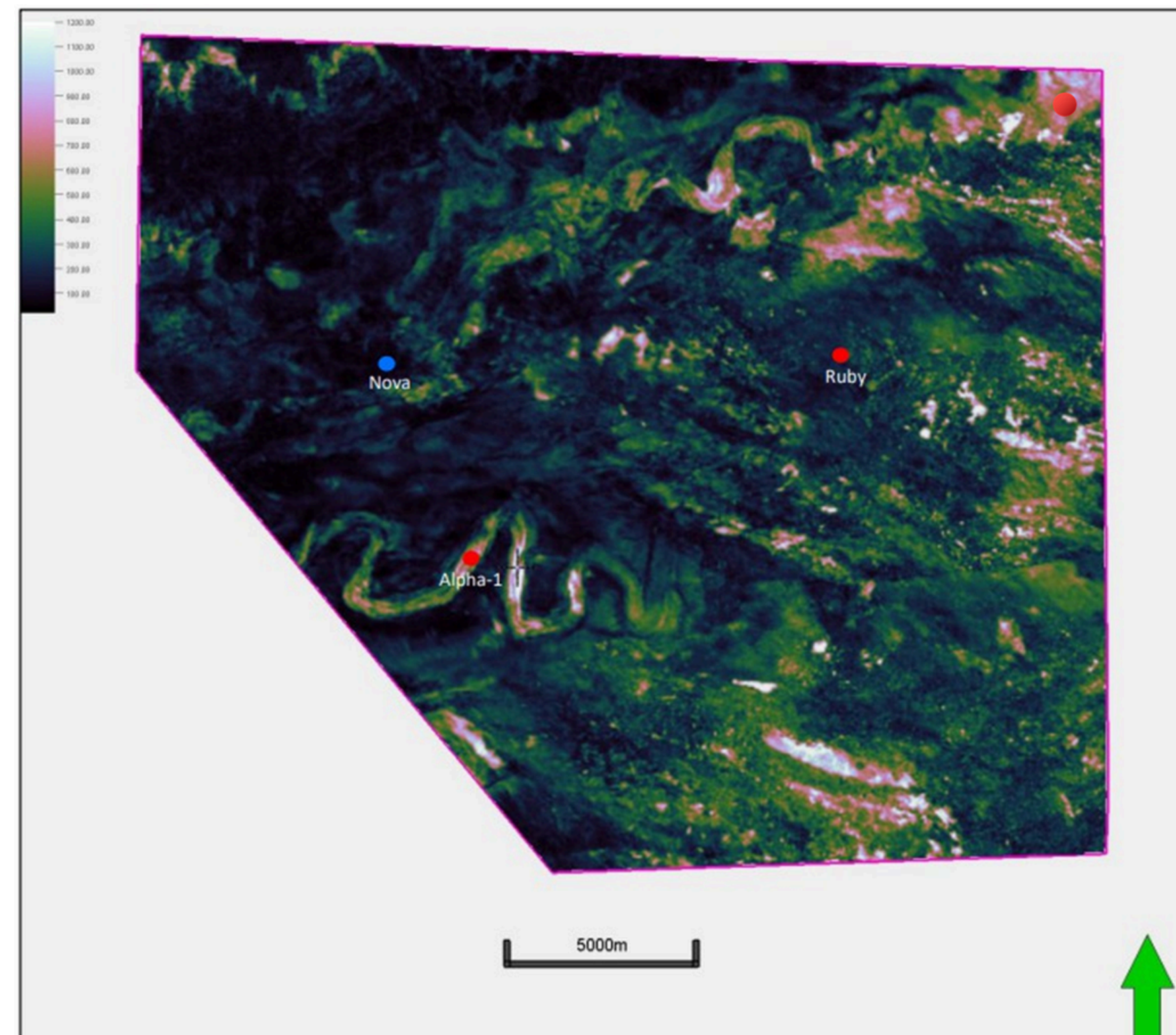
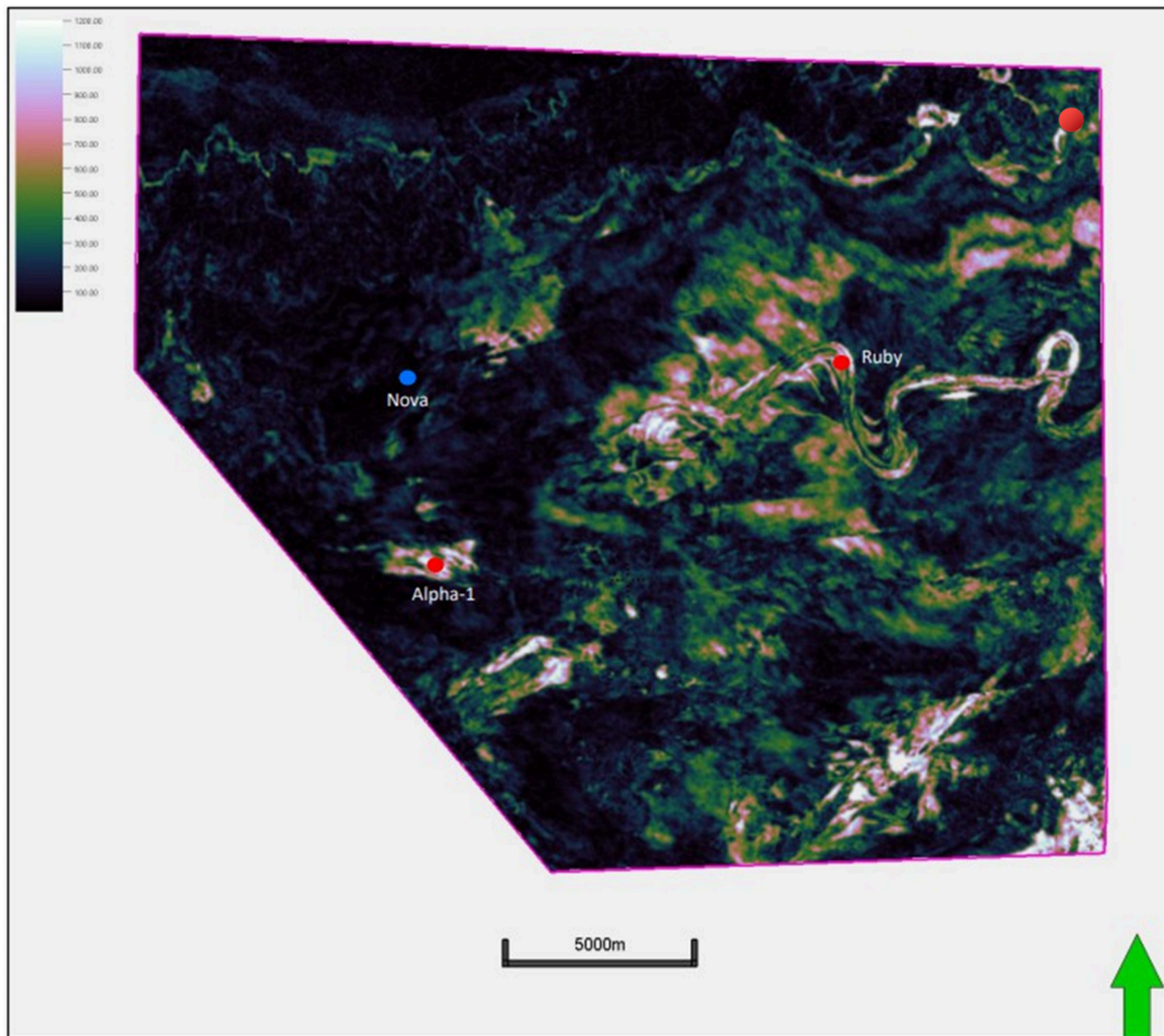


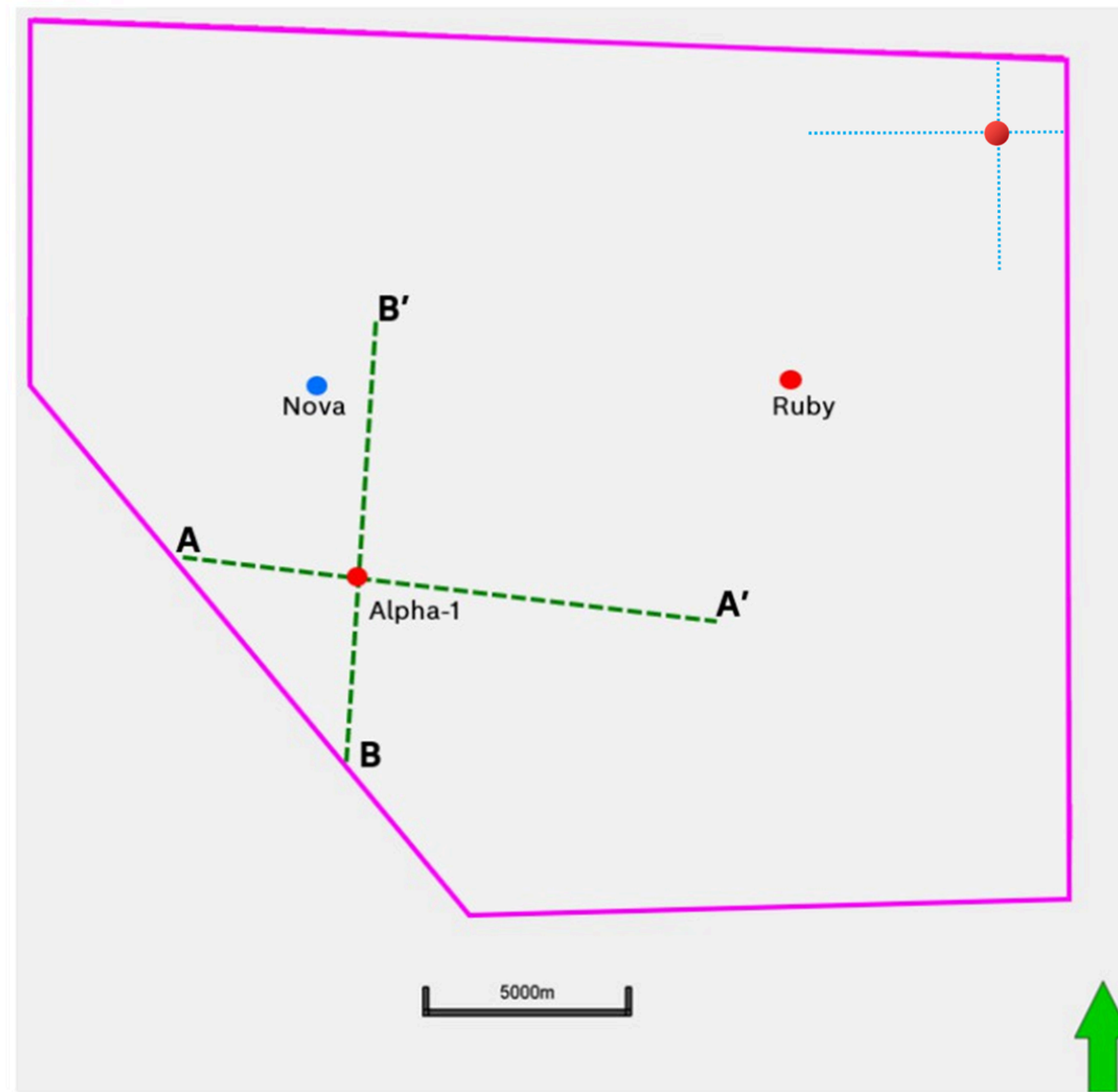
Appraisal Well



New opportunity in the Block

- Identify new leads(if any) in the block. Also think of additional targets/ leads in the vicinity of Alpha-1 o Provide justification and description of the leads using petroleum system elements, well analysis and other information in the pack.
- Near Alpha, there is a high chance of occurrence as indicated by the RMS seismic attribute, seismic sections, and well log information. Due to their close proximity, seismic analysis can be performed to gain a better understanding of areas closer to Alpha.
- Seismic RMS attribute analysis shows channel-like features with relatively high impedance, which are interpreted as channelized depositional systems. Based on the available seismic data and geological understanding, it is likely that stream flow existed between the Alpha and Ruby areas, leading to the deposition of sediments within these channels.
- These sediment deposits indicate the possible presence of reservoir-quality sands, along with a deeper source rock and the requirement for an effective seal. Although the petroleum system is still uncertain, the integration of seismic attributes and geological interpretation suggests a reasonable chance of hydrocarbon presence in the leading wells.
- To increase confidence in the geological interpretation and petroleum system assessment, additional RMS attribute analysis and more seismic sections are required. In particular, seismic sections along the C–C' and D–D' directions for Sol, as well as E–E' and F–F', would be highly useful in confirming channel continuity,
- trap geometry, and identifying new well locations.





Riskiest element of lead.

- Reservoir quality and the thickness of the reservoir is the riskiest element, because there is lateral heterogeneity

Key uncertainties in the block.

- The main uncertainty in the block is related to structure and stratigraphy, as the exact geological framework and petroleum system of the area are not completely understood. The interpretation is largely based on seismic data, which has limited resolution and can lead to multiple geological interpretations.
 - Although bright spots are observed on seismic attributes, they do not always indicate the presence of hydrocarbons. These anomalies may also be caused by lithological changes or tuning effects, which introduces uncertainty in hydrocarbon prediction.
 - There is also uncertainty regarding the presence and thickness of the reservoir. Away from the existing well control, it is not clear whether good-quality reservoir sands are present or how thick the hydrocarbon-bearing zone may be.
 - Trap integrity is another uncertainty, particularly related to fault sealing. It is not fully known whether the faults act as effective seals or allow hydrocarbons to leak, which directly affects hydrocarbon accumulation.
 - Uncertainty also exists in hydrocarbon migration and charge. The effectiveness and timing of hydrocarbon migration from the source rock to the reservoir are not well constrained.
 - Additionally, the hydrocarbon phase (gas, oil, or mixed) and the exact position of fluid contacts are uncertain due to limited pressure and fluid data.
 - Finally, reservoir quality away from the drilled wells is uncertain, as porosity and permeability may vary laterally. Additional seismic and well data would be required to reduce these uncertainties and improve confidence in drilling new leads.
- are not certain that the stratigraphic structure which we had

• **RISK & UNCERTAINTY ASSESSMENT**

Geological Chance of Success (COS)

Source Rock (Low Risk)

- Deep marine source rocks are proven regionally Gas charge confirmed by DST and MDT Thermogenic gas expected at depth
- Probability = 0.95

Migration (Low Risk)

- Faults act as vertical migration pathways
- Gas present in Alpha-1 and Ruby confirms effective migration
- Flat spot observed on seismic supports charge
- Probability = 0.90

Reservoir (Moderate Risk)

- **Good porosity (0.20–0.25) observed in Alpha-1**
- **Turbidite sands show lateral variability**
- **Net pay uncertainty exists away from well control**
- **Probability = 0.70**

Trap (Moderate Risk)

- Combination structural–stratigraphic trap Fault seal effectiveness uncertain
- Trap geometry clear on PSDM seismic
- Probability = 0.75

Seal (Moderate-Low Risk)

- Thick marine shale provides regional top seal Minor risk of fault leakage
- No evidence of leakage in Alpha-1
- Probability = 0.80

$$\text{COS} = P_{\text{source}} \times P_{\text{migration}} \times P_{\text{reservoir}} \times P_{\text{trap}} \times P_{\text{seal}}$$

$$\text{COS} = 0.95 \times 0.90 \times 0.80 \times 0.75 \times 0.70 = 0.36 \text{ (36\%)} \cdot$$

MODERATE – VIABLE

Uncertainty in our interpretation & Additional information/log data that could improve confidence

- Biogenic HC probability is very uncertain due to lack of well log data.
- Migration direction.
- Thermogenic HC: it can be mixed HC.
- Needs gas log, sonic log, and geochemical gradient log.
- Cap rock above reservoir:
- Location and its lithology.
- Type of seal assumed shale, but cannot be determined unless well log is given in that range.
- Jurassic shale and Pliocene shale probability is also uncertain due to lack of data at that depth.
- Growth fault roll-over is bit uncertain.
- It can be an anticline structure or salt dome (less likely), but it is passive margin, so growth fault probability is high.
- Channeling trap may have hydrocarbon or not.
- It is also uncertain unless well log is given in that range.

Uncertainty in our interpretation & Additional information/log data that could improve confidence

- All fault estimation near the reservoir and in seismic is because of low resolution of data and human errors.
- Folds and traps are also uncertain.
- Appraisals are uncertain unless we have more seismic sections of the wells.
- But we are estimating based on given seismic attributes at two different depths, purely on intuition and channels and high impedance.
- New leads are uncertain unless we have seismic data and more well log data of Ruby and seismic of Luna.
- Seismic section of Luna–Ruby.
- Seismic section of CC' and DD', and more well log data.
- Seismic section of Luna in EE' and FF'.
- We also need seismic section of Ruby for less uncertainty of appraisals to be drilled.

Thank You

GeoGliders Team

Naba · Sparsh · Piryanka Yadav · Subham · Pushkar

bp Integrated Subsurface Case Study · Mustang Basin